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PAPER

The role of trust in public acceptability of energy projects: Integrity versus competence

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Abstract

Public acceptability of energy projects depends on people's trust in agents responsible for those projects. We examined to what extent different dimensions of trust, notably integrity- and competence-based trust, are associated with public acceptability of real ongoing energy projects associated with acute risks and other consequences. A series of questionnaire studies in the Netherlands revealed that both integrity- and competence-based trust were positively associated with public acceptability of natural gas extraction which causes earthquakes in the region. Yet, integrity-based trust was more strongly and consistently associated with public acceptability of the natural gas extraction compared to competence-based trust. The findings were rather consistent across four different measurement phases. We discuss the theoretical and practical implications of our findings.

1. Introduction

A stable production and provision of energy is essential to social development and human welfare [1]. The successful implementation and continuation of energy projects depends on the extent to which the public finds the projects acceptable [2–5]. We define public acceptability as the extent to which people evaluate the energy projects (un) favourably. Indeed, low public acceptability can lead to delays and even cancelations of energy projects [6–8].

Energy projects are highly complex [9, 10] and the general public typically does not have sufficient knowledge and skills to implement and manage those projects, and thus has to rely on other agents to do so. Consequentially, trust in agents that are responsible for implementing and managing energy projects, such as governments and energy companies, is an important factor for public acceptability [11–16]. The literature on social license to operate also shows that trust is important for enhancing public acceptability and social approval of energy projects, because people are more likely to evaluate decisions made by responsible agents as legitimate and credible when they have higher trust in the agent [17, 18].

Trust in responsible agents is a complex, much encompassing concept, which has different dimensions (e.g., see [19] for a review). We will focus on two dimensions of trust that have been found to be related to public acceptability, namely competence-based trust and integrity-based trust [4, 11, 15, 20, 21]. Specifically, people need to trust that these agents have the knowledge and skills to implement and manage the project (i.e., competence-based trust). In addition, people need to trust that the agent is open and honest about managing the project and takes public interests, such as their safety, into account (i.e., integrity-based trust). Research shows that both stronger competence- and integrity-based trust are related to higher public acceptability of energy projects [4, 11, 15, 20, 21]. For example, people found a carbon dioxide capture and storage project more acceptable when they thought that the responsible agent has relevant knowledge and skills to implement and manage the project (i.e., high competence-based trust; [15]). Similarly, the more people perceived the responsible agent as caring about public interests (i.e., high integrity-based trust), the more acceptable they found energy projects proposed and managed by the agent, such as climate engineering [11, 20, 21].

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An important question that remains, however, is to what extent each of the two dimensions of trust is important for public acceptability of energy projects. In other words, are both dimensions of trust equally important for acceptability of energy projects, or is one dimension of trust more important than the other for acceptability of energy projects ? Such knowledge would reveal which dimension of trust to prioritise when designing energy projects and which one could be the main barrier for public acceptability. So far, the understanding of this question is limited, as studies typically considered how integrity- or competence-based trust relate to public acceptability of energy projects separately. To study both dimensions of trust at the same time is of particular importance, because integrity-based trust and competence-based trust might be correlated [22, 23]. This could mean that integrity-based and competence-based trust do not both uniquely predict acceptability of the energy project as one influences the other, rather than both being important factors for acceptability of an energy project. Hence, the question remains whether and to what extent integrity-based trust and competence-based trust are uniquely related to acceptability of an energy project when controlling for the other dimension of trust.

Correlational studies on real-life genetically modified field projects suggest that perceived morality of responsible agents tends to be more strongly related to project acceptability than perceived performance of responsible agents [21]. Arguably, this is because perceived morality is related to an agent's good or bad intentions about public interests (i.e., integrity), which can particularly influence the extent to which the public finds the project acceptable. In contrast, perceived performance is less indicative of the good or bad intentions of responsible agents, but more of whether the agent has sufficient knowledge and skills to manage the project (i.e., competence). Thereby, competence-based trust is likely to be less useful for inferring whether public interests will be secured, and thus less relevant for evaluating how acceptable people find the project (cf. [24, 25]). The stronger importance of an agent's morality compared to its competence is also rooted in evolutionary psychology. Namely, knowing whether an agent has good or bad intentions regarding harming or protecting you (i.e., integrity) is more important for survival than knowing whether the agent has the knowledge and skills (i.e., performance) to act in line with their good or bad intentions [26]. Drawing on these strings of literature, we expect that integrity-based trust, which primarily relates to the morality of the responsible agent, would have a stronger effect on acceptability of energy projects than competence-based trust, which primarily relates to the performance of the agent.

To our best knowledge, only one study on the acceptability of energy projects examined both dimensions of trust so far, providing initial evidence that integrity-based trust might be more important for public acceptability than competence-based trust [4]. Specifically, people found a wind energy project more acceptable when they trusted that the responsible agent is honest and open about its activities and considers public interests (i.e., integrity-based trust), irrespective of the extent to which they thought the agent has sufficient competence to implement and manage the project (competence-based trust). Whereas higher competence-based trust only led to higher project acceptability when integrity-based trust was low [4]. In this study, however, people evaluated a hypothetical energy project, with a hypothetical responsible agent, so the question remains whether similar results would be found for a real ongoing energy project, where people are exposed to actual risks and other consequences of the energy project. Research suggests that people may evaluate hypothetical and real energy projects differently [27]. For example, people tend to evaluate the responsible agents and the project more positively when they are hypothetical rather than real, as they do not experience real threats from the project [27], which could potentially influence the relationship between trust and acceptability.

1.1. Current study

As a case in point, we study public acceptability of an ongoing natural gas extraction project in the province of Groningen, the Netherlands, and people's trust in the main responsible agent[2] for this project, namely the Dutch Petroleum Company Nederlandse Aardolie Maatschappij (NAM). Natural gas is an important energy source in the Netherlands as it is the primary energy source for heating houses and water, and for cooking. However, natural gas extraction causes earthquakes in the province of Groningen. The strongest earthquake so far was 3.6 on the Richter scale, namely the Huizinge earthquake in August 2012. Inhabitants of the province of Groningen face various negative consequences caused by the earthquakes due to the natural gas extraction, such as damage to houses and drop in house values, physical injury, stress and worry, and reduced quality of living. NAM operates the natural gas extraction and is responsible for assessing, preventing, and mitigating the risks associated with the natural gas extraction. We aim to study to what extent integrity-based and competence-based trust in the NAM are associated with how acceptable the public finds the ongoing natural gas extraction that comes with acute risks of earthquakes and other consequences.

This study is part of a large-scale longitudinal questionnaire study on public opinion about the natural gas extraction and the induced earthquakes in the province of Groningen, consisting of six waves carried out

[2] The Dutch government is also an important responsible agent in this project as they decide on how much gas will be extracted (see [28]).

Table 1. Number of participants per research phase.

	Phase 1 November—December 2013	Phase 2 June—July 2014	Phase 3 December 2014	Phase 4 December 2016
Total *N* (new participants)	390	429 (255)	413 (159)	329 (129)

Note. Not all participants completed all items in the questionnaire, resulting in varying sample sizes in our analyses. New participants refer to those who did not take part in any of the previous phases.

between November 2013 - May 2019 (for more details, see [29, 30]). The current paper is based on data from the first four phases of the longitudinal study, as the last two phases did not include measures of integrity- and competence-based trust in the NAM. The longitudinal design enables us to test our hypothesis over time. Notably, many things happened during the course of the study which may have influenced people's trust in the integrity and the competence of the NAM and their acceptability of natural gas extraction. For example, more induced earthquakes took place in the meantime. Also, the NAM took various mitigation measures to limit the risks caused by natural gas extraction, such as reinforcing houses and compensating people for the damage to their houses (for more details see [29, 30]). We test whether the relationships between the two dimensions of trust and public acceptability of natural gas extraction are robust and remain stable despite possible changes in the levels of trust in the integrity and competence of the NAM and public acceptability of the natural gas extraction.

2. Method

2.1. Procedure and participants

Across the four phases, 933 participants answered the questions relevant for the aims of this study. The total number of participants per phase is depicted in table 1. In each phase, participants were asked whether they are willing to participate in subsequent phases, in which case they were approached again in the next phases. At the same time, in each phase, new participants were recruited to ensure similar sample sizes across the phases. The samples are representative of the general population in the province of Groningen (see appendix A for demographic characteristics of the participants in all four phases).

2.2. Measures

2.2.1. Integrity-based trust

Respondents indicated to what extent they agreed with the following three statements regarding NAM: (1) takes the safety of the population into account, (2) is honest about the risks of gas extraction, and (3) is open about the risks of gas extraction [11, 20]. In phase 1, participants were asked to answer those items when considering gas extraction from the Groningen gas field, and in phases 2–4 when considering earthquakes induced by the gas extraction from the Groningen gas field[3]. Responses were given on a 7-points scale, ranging from 1 *totally disagree* to 7 *totally agree*. We computed the mean score of the three items to represent people's trust in the integrity of the NAM; higher scores represent higher levels of integrity-based trust ($\alpha_{T1} = .92$, $\alpha_{T2} = .90$, $\alpha_{T3} = .91$, $\alpha_{T4} = .91$)[4]. Figure 1 shows that people's trust in the integrity of the NAM was generally low (i.e., below the scale mid-point) and gradually decreased across the four phases.

2.2.2. Competence-based trust

Respondents indicated to what extent they agreed with the following three statements regarding the NAM: (1) is competent, (2) has the necessary skills to limit risks of gas extraction, and (3) has the knowledge needed to limit risks of gas extraction [15, 32]. Again, in phase 1, participants were asked to answer those items when considering gas extraction from the Groningen gas field, and in phases 2–4 when considering earthquakes

[3] We improved the questions of trust in the NAM after phase 1 to better reflect what we aimed to measure: evaluating people's level of trust in the NAM's integrity and competence to manage the risks of the energy project - in this case the earthquakes resulting from the gas extraction activities. Since the respondents in this study are local people exposed to the risks of earthquakes due to gas extraction, it is very likely that they already considered the risks of earthquakes when evaluating their trust in the NAM in phase 1 too. Indeed, despite the slightly different ways of phrasing the items, the pattern of results was very consistent across phases. This suggests that our results are rather robust, despite slightly different phrasing of the items. Still, caution is needed when comparing the results of phase 1 with the results of the other phases.

[4] We inspected whether integrity- and competence-based trust can be empirically distinguished, by employing the Multiple Group Method (MGM)—a simple and effective type of confirmatory factor analysis [31]. Results from the MGM showed that integrity-based trust and competence-based trust could be empirically distinguished in all phases (see appendix B). Notably, the items measuring each dimension of trust correlated strongest with the scale they were assigned to (while correcting for correlations between items and subscales in which the items take part).

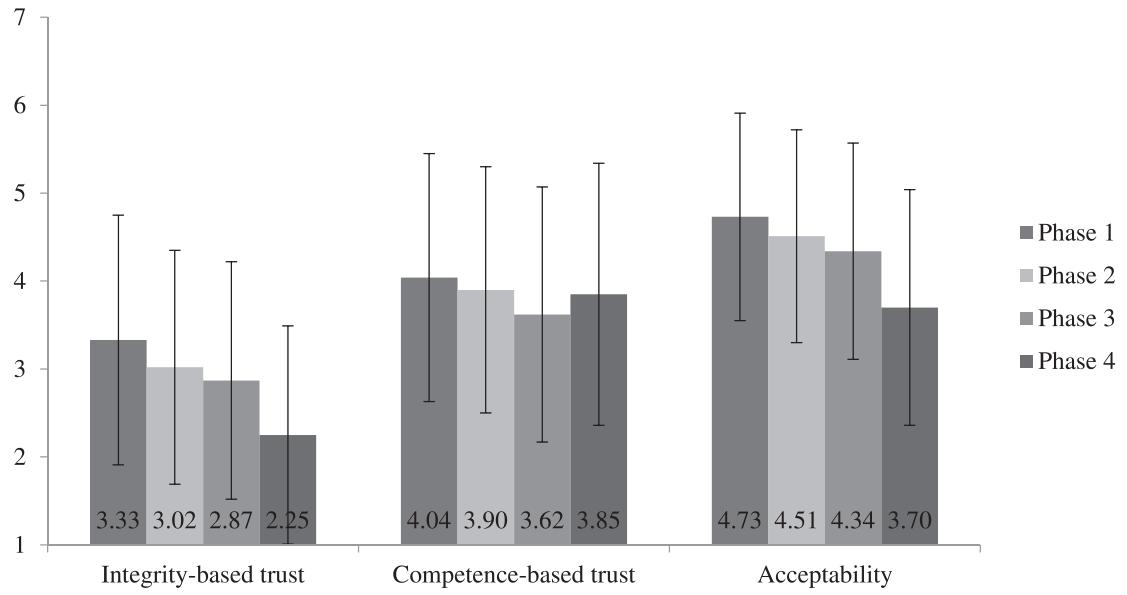

Figure 1. Mean scores and standard deviations of integrity- and competence-based trust in the NAM, and public acceptability of natural gas extraction.

induced by the gas extraction from the Groningen gas field[2]. Responses were given on a 7-points scale, ranging from 1 *totally disagree* to 7 *totally agree*. We computed the mean score of the three items to represent people's trust in the competence of the NAM; higher scores represent higher levels of competence-based trust ($\alpha_{T1} = .88, \alpha_{T2} = .90, \alpha_{T3} = .89, \alpha_{T4} = .83$)[3]. Figure 1 shows that people's trust in the competence of the NAM was not very high (i.e., just above the scale mid-point) and varied only slightly across the phases, from just above to just below the mid-point of the scale.

2.2.3. Acceptability of natural gas extraction

We included four separate items to measure acceptability, namely respondents indicated to what extent they find the natural gas extraction at the Groningen gas field (1) acceptable, (2) useful, (3) good and (4) necessary, respectively, on a 7-point scale ranging from 1 *totally not* to 7 *totally* [4, 33]. We computed the mean score of the four items that reflect the extent to which people find the natural gas extraction acceptable; higher scores represent higher levels of acceptability ($\alpha_{T1} = .83, \alpha_{T2} = .88, \alpha_{T3} = .89, \alpha_{T4} = .90$). Figure 1 shows that people initially found the natural gas extraction somewhat acceptable, yet acceptability decreased across the four phases to just below the scale mid-point.

3. Results

To test the relationship between integrity-based trust, competence-based trust and acceptability of natural gas extraction, we first conducted correlational analysis. Next, we conducted multilevel modelling for repeated measures to study the unique relationship between the two dimensions of trust and acceptability.

3.1. Relationship between integrity-based trust and competence-based trust and acceptability of natural gas extraction

First, we inspected Pearson correlations between integrity- and competence-based trust and public acceptability of natural gas extraction in each phase (see table 2). Both integrity- and competence-based trust were positively and significantly associated with public acceptability: the more people trust that the NAM was integer and competent, respectively, the more they find the natural gas extraction acceptable. As expected, in most phases, integrity-based trust was more strongly correlated with public acceptability, compared to competence-based trust, except in phase 2 where both dimensions of trust were similarly strongly correlated with public acceptability. The relationships were robust across the four phases.

Next, we used multilevel modelling for repeated measures (e.g., [34]) to assess the unique relationships between the two dimensions of trust and public acceptability. This enables us to test the robustness of the relationships while accounting for possible differences across measurement phases [34]. Multilevel analysis allows us to also use responses of respondents that did not participate in all phases, as multilevel analysis is able to

Table 2. Correlations between integrity- and competence-based trust, and acceptability.

		Competence-based trust	Acceptability
Integrity-based trust	Phase 1	.57[a]	.32[a]
	Phase 2	.58[a]	.31[a]
	Phase 3	.55[a]	.40[a]
	Phase 4	.41[a]	.39[a]
Competence-based trust	Phase 1		.23[a]
	Phase 2		.29[a]
	Phase 3		.33[a]
	Phase 4		.27[a]

[a] $p < .001$

Table 3. Results of multilevel regression models testing the relationships between integrity-, competence-based trust and acceptability across time.

	Estimate	*SE*	*p*	95% CI
Fixed effects				
Intercept	4.680	.050	<.001	[4.580; 4.780]
Measurement phase	−.228	.025	<.001	[−.278; −.178]
Integrity-based trust	.145	.039	<.001	[.067; .223]
Competence-based trust	.082	.039	.036	[.004; .160]
Phase x integrity-based trust	.073	.020	<.001	[.033; .113]
Phase x competence-based trust	.002	.019	.916	[−.036; .040]
Random effects				
Level-2 variance τ_0^2	.676	.059		
Level-1 variance σ^2	.674	.038		
R^2	19.016			
Deviance	4528.849			

Note. Measurement phase was coded 0 = 1st phase, ascending.

account for missing data. As such, the multilevel analysis for repeated measures allows for variations in the number of available measurements per respondent. Specifically, multilevel analysis treats each measurement as a different observation of the same variable and omits missing measurements from the analysis assuming they are missing at random (see [35]). We used MLwiN (Version 3.05) to build two-level models where the measurement phases (level 1) were nested within individuals (level 2). Measurement phase was added as an uncentered predictor, whereas the dimensions of trust were added using grand mean centering. Effects were tested with approximate t-tests, applying a significance level of $\alpha = .05$.

Results of the multilevel model are shown in table 3. Measurement phase had a significant negative effect on public acceptability, indicating that public acceptability significantly decreased over time. In line with the correlation analysis, both integrity- and competence-based trust were positively associated with public acceptability. As expected, higher integrity-based trust was more strongly associated with higher public acceptability, compared to competence-based trust. Interestingly, the interaction effect of measurement phase and integrity-based trust was positive and significant, meaning that the effect of integrity-based trust on public acceptability increased over time. The interaction effect of measurement phase and competence-based trust was not significant.

4. Conclusion and discussion

We studied the extent to which integrity- and competence-based trust are associated with public acceptability of a real ongoing energy project with acute risks and other consequences, namely local natural gas extraction. Particularly, we hypothesized that integrity-based trust, compared to competence-based trust, would be more strongly associated with public acceptability. We tested the relationships at different points in time to examine the robustness of the relationships, irrespective of possible changes in the means of the two dimensions of trust and acceptability judgements.

As expected, the results showed that the more people trust that the responsible agent has integrity, and the more they trust that the responsible agent is competent in managing the project, the more the energy project was found acceptable by people. Moreover, as expected, trust in the integrity of the responsible agent was more strongly associated with public acceptability of the natural gas extraction, compared to competence-based trust.

The pattern of results was rather consistent across the different measurement phases, even though public acceptability of the natural gas extraction and the level of integrity-based trust in the NAM decreased over time. This suggests that the relationships we found between the dimensions of trust and public acceptability of the natural gas extraction are rather robust (caution is needed when comparing phase 1; see Footnote 2). Interestingly, the relationship between integrity-based trust and acceptability of the natural gas extraction got even stronger over time. The NAM changed some practices in limiting risks caused by the gas extraction over the years which may have influenced the effect of integrity-based trust on the acceptability of natural gas extraction. We did not measure how people perceived such changes and thus we could not control for these perceived changes empirically. Importantly, the main aim of this paper is to examine the relationship between the two dimensions of trust and acceptability, and we found integrity-based trust was more strongly associated with acceptability than competence-based trust across different measurement phases.

Our results have important theoretical implications. The findings are in line with the literature which shows that in situations where people have to rely on others, two factors are important for people's responses to such situations: people's perception of the other's good intentions (i.e., their integrity), and their knowledge and skills (i.e., their competence; cf [24–26, 36–38]). More importantly, the finding that integrity-based trust is more important for acceptability of energy projects compared to competence-based trust, not only applies for hypothetical projects with hypothetical agents where people may not yet experience and/or foreseen actual acute risks and other consequences [4], but also applies for real ongoing energy projects with actual acute risks and other consequences. These findings support our reasoning that trusting whether an agent will be open and honest and protect public interests, which is reflected in integrity-based trust, is more important for public acceptability of an energy project than trusting whether an agent has the knowledge and skills to manage the project. Future research could test whether integrity-based trust also plays a more important role in acceptability in different contexts, such as with different types of operators involved (e.g., government versus industry), and for different types of energy projects (e.g., projects that may involve different costs and benefits), and culture.

Our findings also have important practical implications. Many energy companies are trying to improve people's trust in them by emphasising that they are open and honest and care about public interests through promotions in the media (i.e., high in integrity-based trust) [39], and that they are competent in managing their energy projects through recruiting staff with sufficient knowledge and skills (i.e., high in competence-based trust) [40, 41]. Our results suggest that considering both dimensions of trust is important for securing public acceptability. Yet, strategies focused on integrity-based trust seem particularly important to enhance and secure public acceptability of energy projects, as our results suggest that (perceived) integrity of energy companies might be more important for public acceptability of their energy projects than (perceived) competence of energy companies. Hence, in order to enhance public acceptability of the energy projects, energy companies primarily need to be open and honest about the energy projects and the associated risks and other consequences, and protect public interests, which could be one way to enhance people's integrity-based trust. Importantly, it may not be sufficient for energy companies to only communicate that they care about public interests. Rather, their actions need to demonstrate that they have integrity. If they say they are honest but cannot prove so or even worse that their actions show the opposite, this may even lower people's trust in their integrity. Future research could study what are other effective ways to enhance trust besides being open and honest about risks and protecting public interests, such as organizing more transparent public participation and decision-making procedures to increase integrity-based trust.

Due to the correlational design of our study, it cannot be ruled out that causality can also go in the other direction, namely that acceptability influences the extent to which people trust the responsible agent. While the literature shows that indeed having higher integrity-based and competence-based trust enhances acceptability of energy projects, it could also be that people trust that the agent has integrity and is competent, because they already think the project is of good quality and find the project acceptable. Future research could apply (field) experimental designs or longitudinal designs to study if this can be the case.

Future research could test if and how integrity-based trust and competence-based trust are related to each other and whether one influences the other. We found that the two dimensions of trust are positively related to each other (see table 2). This is in line with other studies showing that the two dimensions of trust are positively correlated [22]. Following the social cognition literature, we assumed that integrity and competence are two conceptually different components, which people rely on to form an evaluation of an agent [26, 37, 42], and our findings support this assumption as we were able to empirically distinguish the two dimensions of trust and found that both dimensions were uniquely associated with public acceptability. Other studies also found that integrity- and competence-based trust can be empirically distinguished (e.g., [21]). Future studies could consider other dimensions of trust as well, such as

trust in past behaviour of responsible agents [43], and test how it is related to public acceptability, next to integrity-based and competence-based trust.

To conclude, our research empirically examined how integrity- and competence-based trust are related to public acceptability of a real ongoing energy project associated with acute risks and other consequences. Our study indicates that both higher integrity- and competence-based trust are associated with higher public acceptability of real energy projects. Yet, integrity-based trust was most strongly associated with public acceptability of the energy project compared to competence-based trust. The pattern of the results was rather consistent over four different measurement phases. Policies aimed at promoting public acceptability of energy projects should particularly consider increasing perceived integrity of the responsible agent(s), as to secure public acceptability and thereby a stable energy production and provision.

Data availability statement

The data that support the findings of this study are available upon reasonable request from the authors.

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Appendix A

Table A1. Socio-demographic characteristics of the participants per research phase

		Phase 1	Phase 2	Phase 3	Phase 4
Gender	Female	43%	45%	43%	46%
	Male	54%	52%	56%	53%
	Missing values	2%	2%	1%	1%
Age (in years)	*M (SD)*	52.15 (14.91	52.66 (14.59)	54.81 (13.71)	51.37
(15.54) (15.54)	Minimum	19	18	20	18
	Maximum	90	84	84	91
	Missing values	3%	2%	1%	1%
Highest completed education	Primary school	2%	1%	1%	5%
	Lower vocational education	9%	9%	7%	8%
	Secondary (vocational) education	36%	33%	35%	33%
	Higher (vocational) education	36%	31%	27%	41%
	Scientific education (university)	12%	9%	6%	10%
	Other	2%	1%	1%	0%
	Missing values	3%	16%	22%	3%
Income per month	< €1000	4%	3%	3%	7%
	€1000-€2000	25%	26%	23%	23%
	€2000-€3000	28%	25%	23%	25%
	€3000-€4000	20%	17%	18%	30%
	€4000-€5000	8%	6%	8%	5%
	>€5000	4%	2%	1%	5%
	Missing values	11%	21%	25%	7%
Length of residence in the area (in years)	*M (SD)*	30.59 (19.66)	31.95 (19.3	34.40 (19.63)	32.25 (19.66)
	Minimum	.50	1	1	1
	Maximum	83	83	80	83
	Missing values	2%	15%	20%	1%

Appendix B

Table B1. Multiple Group Method (MGM) for integrity- and competence-based trust.

		Phase 1		Phase 2		Phase 3		Phase 4	
		Integrity-based trust	Competence-based trust	Integrity-based trust	Competence-based trust	Integrity-based trust	Competence-based trust	Integrity-based trust	Competence-based trust
Integrity-based trust	The NAM takes the safety of the population into account.	**.74**	.46	**.71**	.53	**.71**	.46	**.75**	.35
	The NAM is honest about the risks of gas extraction.	**.81**	.48	**.79**	.47	**.80**	.48	**.82**	.31
	The NAM is open about the risks of gas extraction.	**.79**	.48	**.78**	.45	**.79**	.42	**.77**	.29
Competence-based trust	The NAM is competent.	.49	**.62**	.48	**.71**	.46	**.69**	.37	**.54**
	The NAM has the necessary skills to limit risks of gas extraction.	.46	**.75**	.52	**.79**	.49	**.75**	.32	**.65**
	The NAM has the knowledge needed to limit risks of gas extraction.	.46	**.76**	.46	**.79**	.41	**.73**	.27	**.68**

Note. The correlations in the table are corrected for correlations between items and subscales in which the items take part. Strongest correlations are printed in bold.

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